

You can ignite TNT in Adventure mode

MC-222818 Items

Status	Resolved / Fixed
Affected	1.16.5, 21w14a, 21w15a, 21w16a, 21w17a, 21w18a (+31 more)
Fixed in	26.3 Snapshot 1
Reported	2021-04-09

STEPS TO REPRODUCE

1. Launch Minecraft 1.19.2 and create a new Creative world with default settings.
2. Obtain flint and steel.
3. Obtain TNT.
4. Place the TNT in the world.
5. Switch the player into Adventure mode.
6. Use the flint and steel on the placed TNT.
7. Observe that the TNT can be ignited in Adventure mode.

OBSERVED BEHAVIOUR

In Adventure mode, a player can ignite a placed TNT block by using flint and steel on it. The TNT is removed and becomes primed, beginning its normal fuse and subsequently exploding. This occurs even when the player does not have the ability to build in Adventure mode.

The interaction is accepted based solely on the held item being flint and steel or a fire charge; the TNT interaction does not check the player's `mayBuild` ability or otherwise restrict the action for Adventure mode.

EXPECTED BEHAVIOUR

In Adventure mode, attempting to ignite TNT with flint and steel should not activate it or cause an explosion. The TNT should remain in place and unchanged, since Adventure mode should prevent the player from igniting it, consistent with the inability to ignite candles or campfires without the necessary building permissions.

ENVIRONMENT

Minecraft Java Edition Versions: 1.16.5, 21w14a, 21w15a, 21w16a, 21w17a, 21w18a, 21w19a, 21w20a, 1.17 Pre-release 1, 1.17 Pre-release 3, 1.17 Pre-release 4, 1.17 Pre-release 5, 1.17 Release Candidate 1, 1.17 Release Candidate 2, 1.17, 1.17.1 Pre-release 1, 1.17.1 Pre-release 2, 1.17.1 Release Candidate 1, 1.17.1, 21w37a, 21w43a, 1.18, 1.18.1, 22w05a, 22w06a, 1.18.2, 1.19, 1.19.2, 22w43a, 1.19.3, 1.19.4, 1.20.1, 24w20a, 1.21, 1.21.3, 1.21.4, 1.21.7 (includes snapshots)

ORIGINAL DESCRIPTION

The Bug:

You can ignite TNT in adventure mode.

This behavior is inconsistent as the player cannot ignite candles or campfires in adventure mode, so one would expect the same behavior for TNT.

Steps to Reproduce:

- Obtain some flint and steel, place down some TNT, and switch into adventure mode.
- Attempt to ignite the TNT using the flint and steel.
- Take note as to whether or not you can ignite TNT in adventure mode.

Observed Behavior:

TNT can be ignited.

Expected Behavior:

TNT would not be able to be ignited.

Code Analysis:

Code analysis by Avoma can be found below.

The following is based on a decompiled version of Minecraft 1.19.2 using MCP-Reborn.

net.minecraft.world.level.block.TntBlock.java

```
public class TntBlock extends Block {
```

```
...
```

```
public InteractionResult use(BlockState blockState, Level level, BlockPos blockPos, Player player, InteractionHand interactionHand, BlockHitResult blockHitResult) {
```

```
    ItemStack itemstack = player.getItemInHand(interactionHand);
```

```
    if (!itemstack.is(Items.FLINT_AND_STEEL) && !itemstack.is(Items.FIRE_CHARGE)) {
```

```
        return super.use(blockState, level, blockPos, player, interactionHand, blockHitResult);
```

```
    } else {
```

```
        explode(level, blockPos, player);
```

```
        level.setBlock(blockPos, Blocks.AIR.defaultBlockState(), 11);
```

```
        Item item = itemstack.getItem();
```

```
        if (!player.isCreative()) {
```

```
            if (itemstack.is(Items.FLINT_AND_STEEL)) {
```

```
                itemstack.hurtAndBreak(1, player, (player1) -> {
```

```
                    player1.broadcastBreakEvent(interactionHand);
```

```
                });
```

```
            } else {
```

```
                itemstack.shrink(1);
```

```
            }
```

```
        }
```

```
        player.awardStat(Stats.ITEM_USED.get(item));
```

```
        return InteractionResult.sidedSuccess(level.isClientSide);
```

```
    }
```

```
}
```

```
...
```

If we look at the above class, we can see that there is only one necessary check that's carried out before allowing TNT to be ignited. This check is to quite simply see if the player is holding either flint and steel or a fire charge at the time of the interaction. If they are, the TNT will be ignited. The game doesn't check to see what abilities the player possesses (what game mode they are in) before allowing them to ignite TNT, therefore resulting in this problem occurring.

Fix:

Simply altering the appropriate existing "if" statement

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-222818>